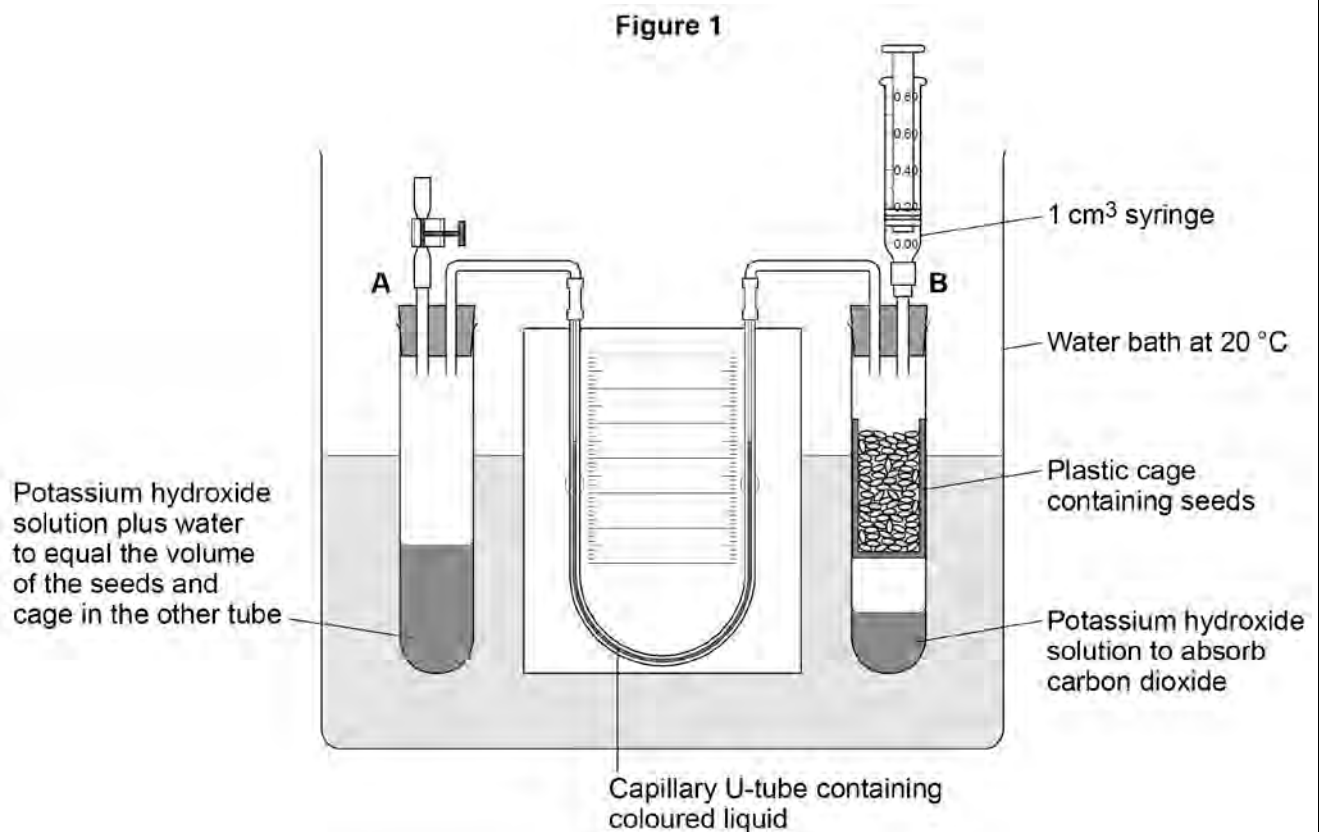


1

Figure 1 shows the apparatus used for measuring the rate of oxygen consumption in aerobic respiration by seeds.



0 1 . 1

For the first 10 minutes, the tap attached to tube **A** was left open and the syringe from tube **B** was removed.

Suggest **three** reasons why the apparatus was left for 10 minutes.

[3 marks]

- 1 _____
- _____
- 2 _____
- _____
- 3 _____
- _____

0 1 . 2 Suggest and explain why the chosen temperature was 20 °C for this experiment. **[2 marks]**

Question 1 continues on the next page

After 10 minutes, the tap attached to tube **A** was closed and the syringe was attached to tube **B**. Every minute, the syringe plunger was moved until the levels in the U-tube were the same. The reading on the syringe volume scale was then recorded.

The results are shown in **Table 1**.

Table 1

Time / minutes	Reading on syringe volume scale / cm ³
0	0.84
1	0.81
2	0.79
3	0.76
4	0.73
5	0.70
6	0.68
7	0.66
8	0.63
9	0.62
10	0.58

0 1 . 3

During the experiment, the coloured liquid in the tubing moved towards tube **B**. Explain what caused this.

[3 marks]

[Extra space]

0	1	.	4
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The mass of the seeds was 1.6 g. Use the information in **Table 1** to calculate the rate of oxygen consumption in $\text{cm}^3 \text{g}^{-1} \text{hour}^{-1}$ by the seeds.

Show your working.

[2 marks]

Rate = _____ $\text{cm}^3 \text{g}^{-1} \text{hour}^{-1}$

Turn over for the next question

